

CRITICALITY ANALYSIS REQUESTS

Fill in the appropriate section(s) below. Each section shows geometry images and has its own defaults.

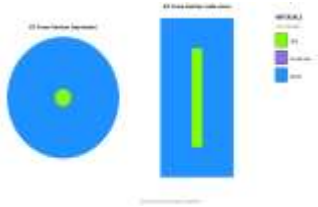
SINGLE CYLINDER

Cold traps, RUTS traps, pump cavities

#	Description	radius_cm	height_cm	enrichment	Notes
1	product cold trap	15-30	50-200	5-24%	
2	product Al2O3 trap	15-30	90-150	5-24%	
3	product NaF trap	15-30	90-150	5-24%	
4	RUTS buffer volume	30-60	30-60	5-24%	
5	Leybold Trivac D658 rotary vane pump internal volume (11.3L)	12 to 15	60-90	5-24%	
6	Leybold RUVAC WS 251 roots pump internal volume (5.64L)	12 to 15	30-60	5-24%	

Defaults Applied:

Wall material	Aluminum, steel + nickel coating inside pump volumes
Wall thickness	0.32 cm (1/8")
Reflector	Water, 30 cm
UF6 density	5.09 g/cc



RECTANGULAR BOX

Chemical traps, HEPA filters, rectangular GEVS

#	Description	length_cm	width_cm	height_cm	enrichment	Notes
1						
2						
3						
4						

Defaults Applied:

Wall material	Steel
Wall thickness	0.32 cm (1/8")
Reflector	Water, 30 cm
UF6 density	5.09 g/cc



PROCESS PIPE

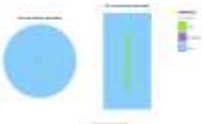
Cascade lines, pigtail - single pipe

pipe size: 1, 1-1/2, 2, 3, 4, 6, 8 (NPS Schedule 10)

#	Description	pipe_size	length_cm	enrichment	Notes
1	product header	NPS 2-3-4	100-2000	5-24%	
2	cassette pigtail	NPS 3/8	100-200	5-24%	
3	cassette manifold	NPS 1-2-3	200-300	5-24%	
4	product cascade row pipeline	NPS 3-4-5	5000-6000	5-24%	

Defaults Applied:

Wall material	Aluminum
Wall thickness	Per ASME B36.10M
Reflector	Water, 30 cm
UF6 density	5.09 g/cc



PARALLEL PIPES

Piping runs, pipe rack spacing studies

num_pipes: 1, 2, or 3 / gap_cm: edge-to-edge spacing

#	Description	num_pipes	pipe_size	length_cm	gap_cm	enrichment	Notes
1	LEU cascade row (one pipe contains feed, one contains product, one pipe contains reflector)	3	NPS 4	5600	2	0.2-0.711-5	T-F-P in each pipe
2	LEU+ cascade row F-P-T (one pipe contains feed, one contains product, one pipe contains reflector)	3	NPS 3-4	5600	2	0.15-0.711-10	T-F-P in each pipe
3	HALEU cascade row F-P-T (one pipe contains feed, one contains product, one pipe contains reflector)	3	NPS 3-4	5600	2	0.711-5-20	T-F-P in each pipe

Defaults Applied:

Wall material	SS304
Wall thickness	Per ASME B36.10M
Reflector	Water, 30 cm
UF6 density	5.09 g/cc



SHIPPING CYLINDER - SINGLE

Single 58, 308, 48Y cylinder

cylinder type: 58, 308, 30Y, 48Y

#	Description	cylinder_type	enrichment	Notes
1	Product cylinder	308	5-24%	HALEU will probably fail
2	Accidental feed cylinder full of product	48Y	5-24%	
3				
4				

Defaults Applied:

Wall material	Per ANSI N14.1 (Monel for 5A/58, carbon steel for larger)
Reflector	Water, 30 cm
Fill fraction	100%
UF6 density	5.09 g/cc



CYLINDER ARRAY - 2D

Custom traps/vessels arranged in rows x cols
gap_cm: edge-to-edge spacing between cylinder walls

#	Description	rows	cols	radius_cm	height_cm	gap_cm	enrichment	Notes
1	RUTS Naf Traps	1	3	35.56	126.5	0-10	5-24%	
2								
3								
4								

Defaults Applied:	
Wall material	Steel
Wall thickness	0.6 cm
Environment	Air, 30 cm boundary
UF6 density	5.09 g/cc



SHIPPING CYLINDER ARRAY - 3D

Stacked shipping cylinders in warehouse storage
orientation: vertical or horizontal / configuration: 2x3x2 (RxCxL) for vertical, 3x2,1 (pattern) for horizontal / environment: air or water

#	Description	cylinder_type	orientation	configuration	gap_yy_cm	gap_xx_cm	environment	enrichment	Notes
1	30B cylinder pad stack	30B	horizontal	3, 2, 1	1	1	air and water	5-24%	
2									
3									
4									

Defaults Applied:	
Wall material	Per ANSI N14.1
Floor	Concrete, 30 cm
Environment boundar	30 cm
UF6 density	5.09 g/cc

